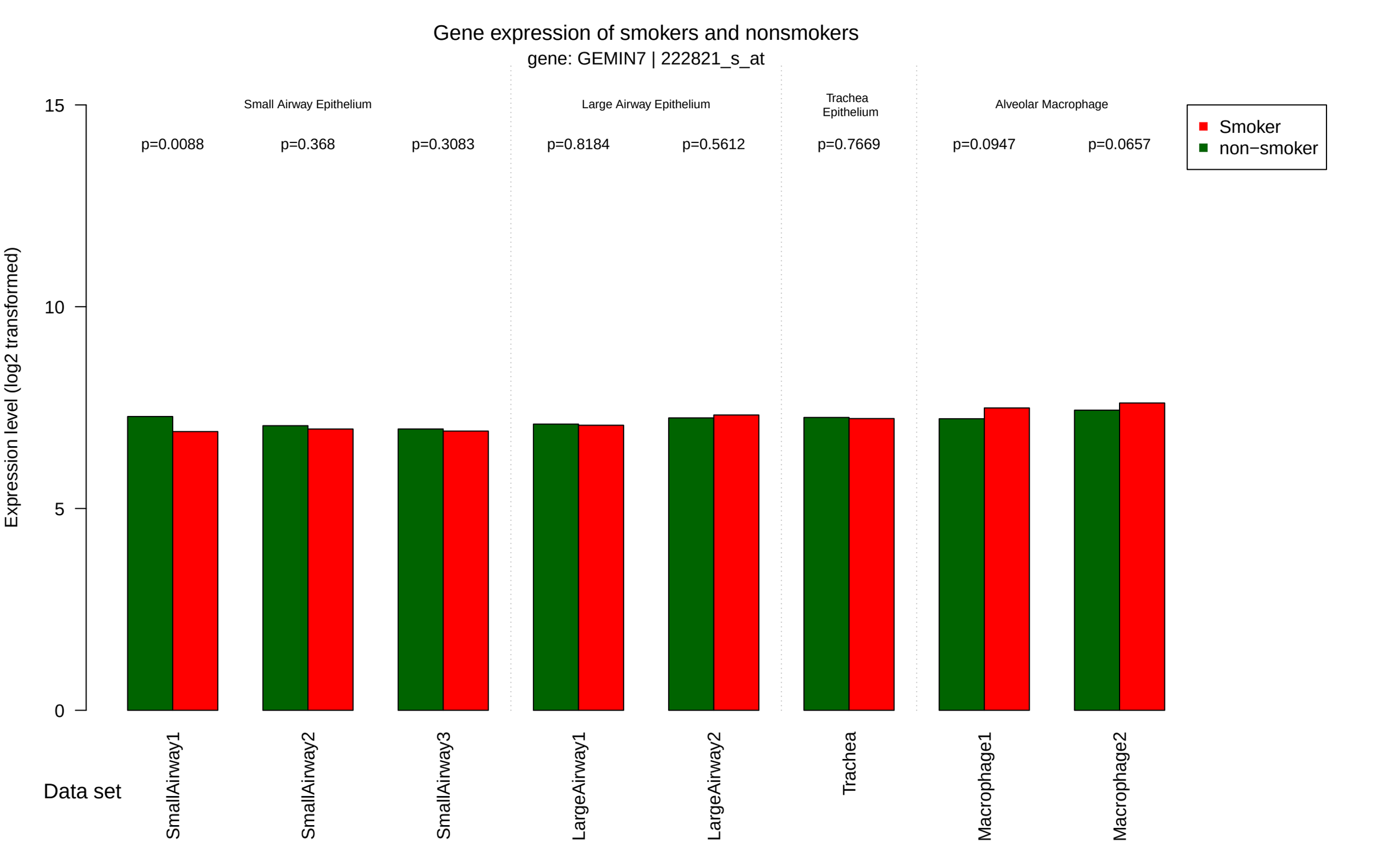
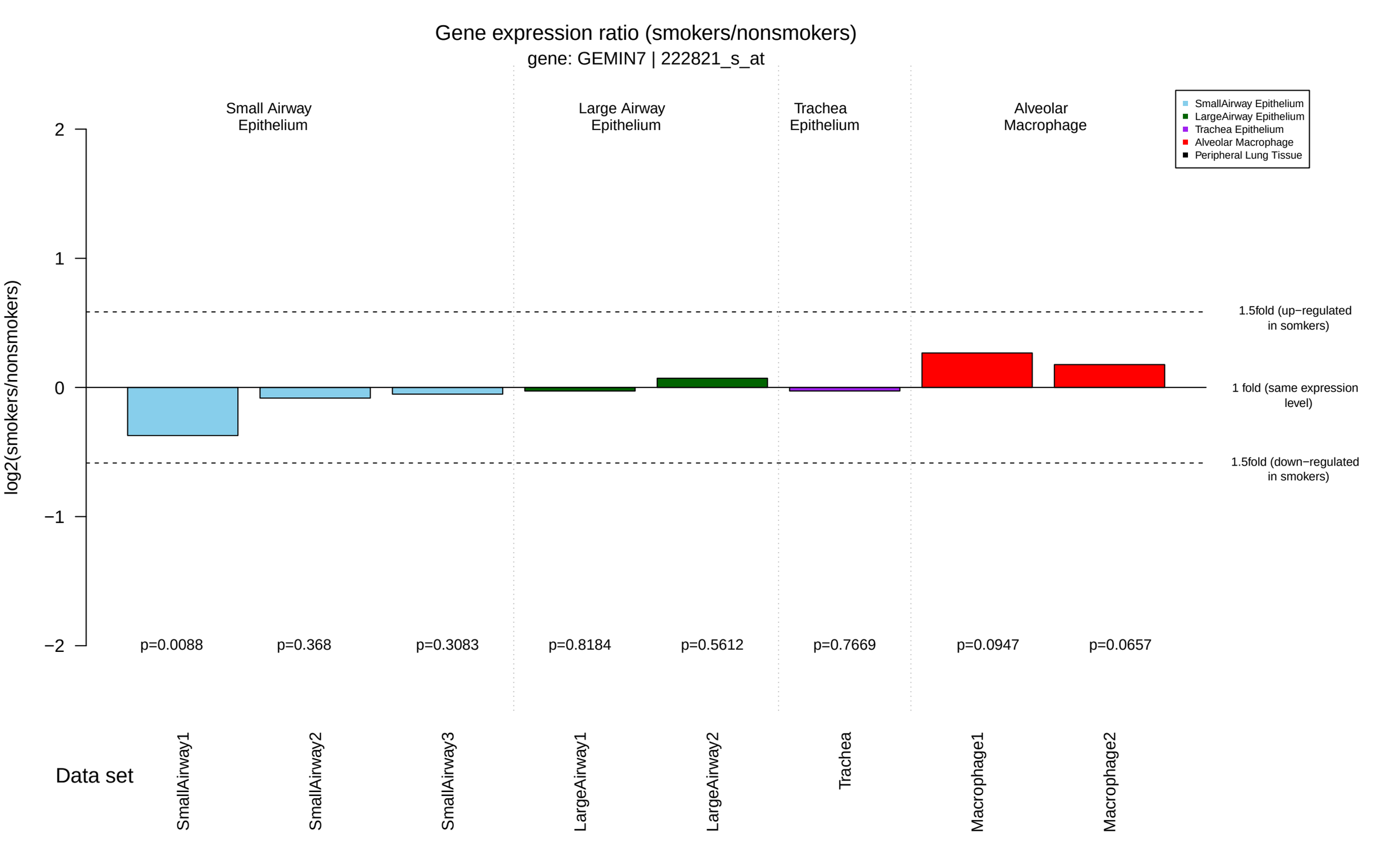
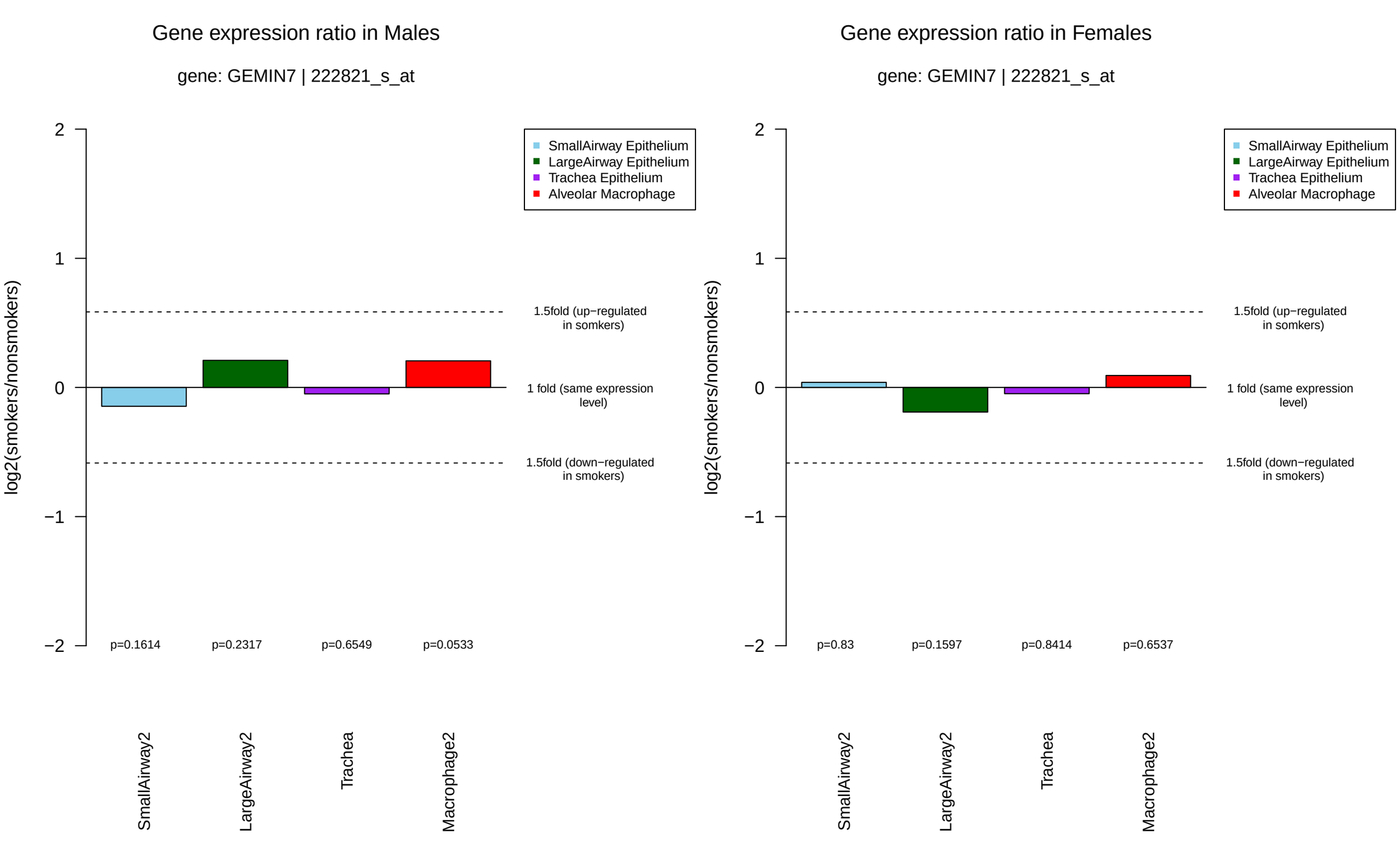
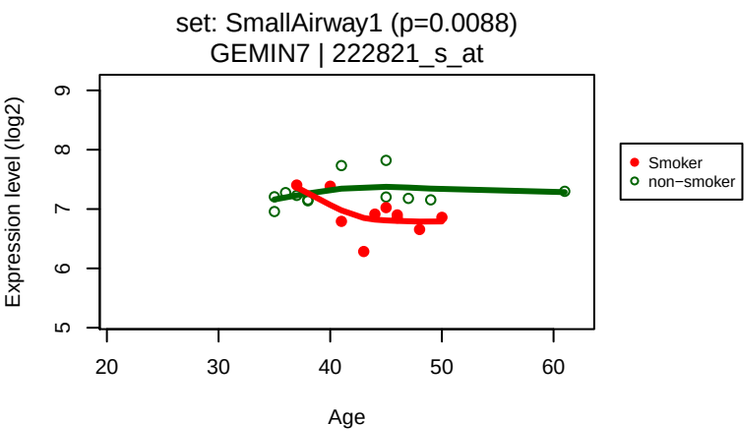
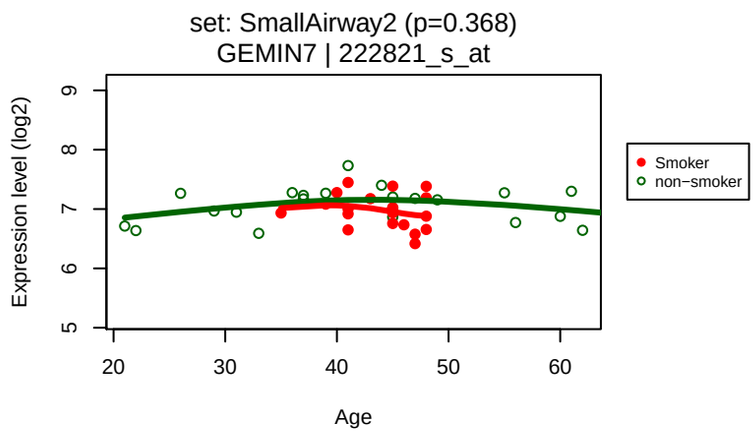
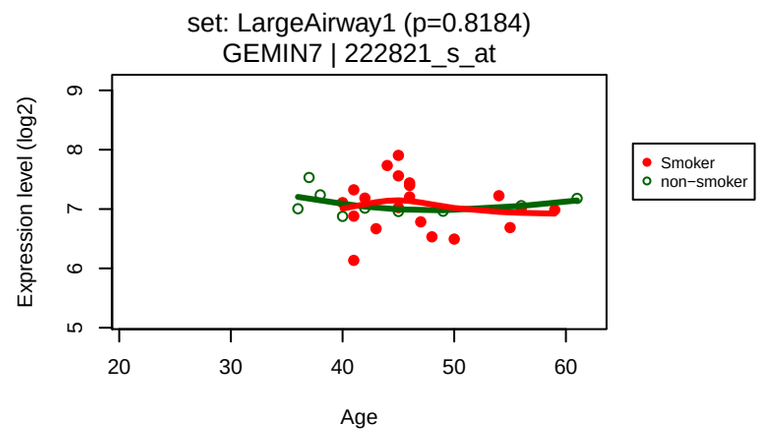

Gene expression of smokers and nonsmokers
gene: GEMIN7 | 222821_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0088
p=0.368
p=0.3083
p=0.8184
p=0.5612
p=0.7669
p=0.0947
p=0.0657
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

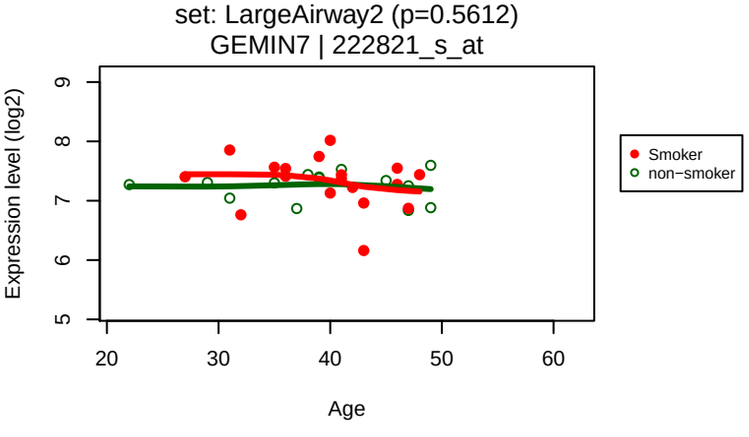

Gene expression ratio (smokers/nonsmokers)
gene: GEMIN7 | 222821_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0088
p=0.368
p=0.3083
p=0.8184
p=0.5612
p=0.7669
p=0.0947
p=0.0657
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

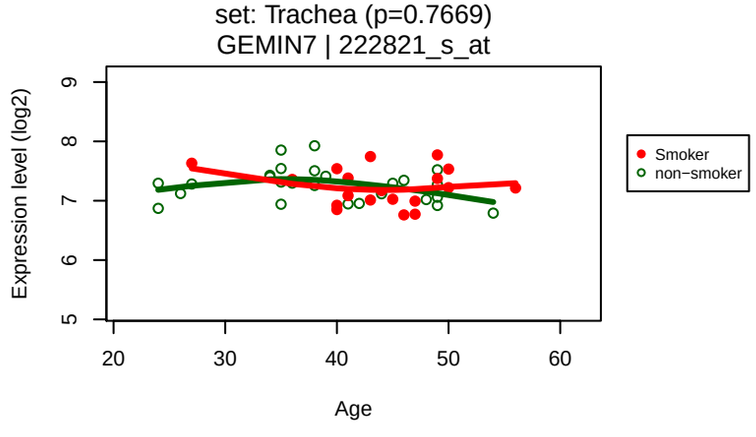

Gene expression ratio in Males
gene: GEMIN7 | 222821_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1614
p=0.2317
p=0.6549
p=0.0533
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GEMIN7 | 222821_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.83
p=0.1597
p=0.8414
p=0.6537
SmallAirway2
LargeAirway2
Trachea
Macrophage2

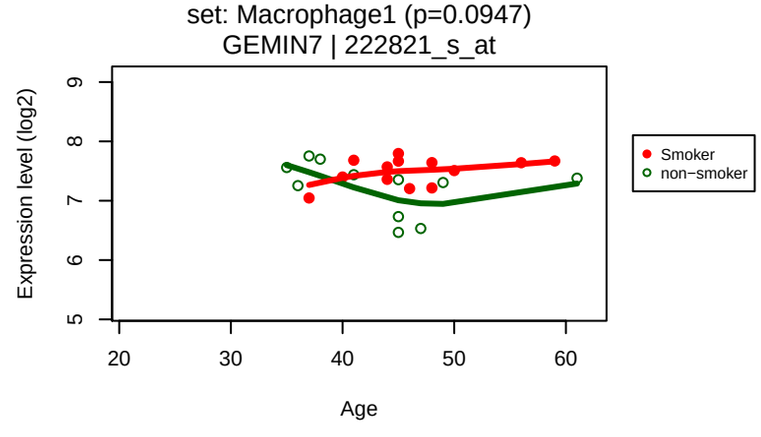

set: SmallAirway1 (p=0.0088)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker

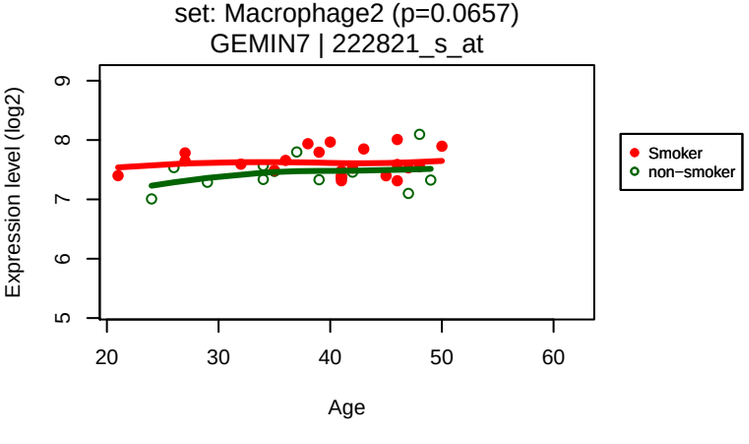

set: SmallAirway2 (p=0.368)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker

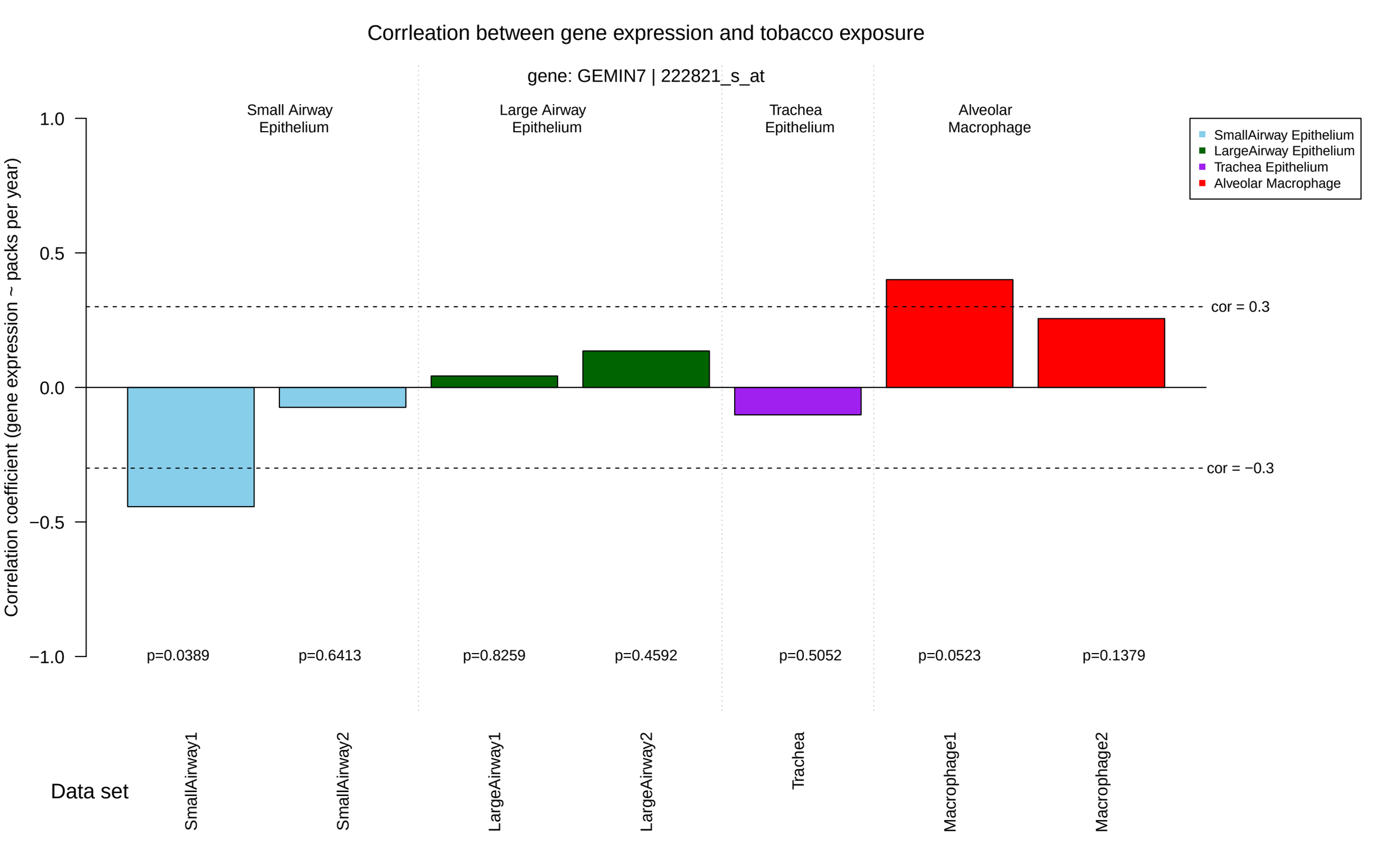

set: LargeAirway1 (p=0.8184)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5612)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7669)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0947)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0657)
GEMIN7 | 222821_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GEMIN7 | 222821_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0389
p=0.6413
p=0.8259
p=0.4592
p=0.5052
p=0.0523
p=0.1379
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2